Bin Hu

Systems Software & Networking | Ph.D. Student, USC

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EDUCATION

University of Southern California (USC)

Ph.D. in Electrical and Computer Engineering | Aug. 2023 - May 2028 (Expected)

Prior doctoral studies: University of Minnesota, CSE, Aug. 2022-Aug. 2023; transferred with advisor.

Southeast University (SEU)

M.Sc. in Computer Science and Engineering | Sept. 2019 - Jun. 2022

B.Eng. in Computer Science and Engineering | Sept. 2015 - Jun. 2019

TECHNICAL SKILLS

Programming: Python, C/C++, Java, MATLAB.

Systems & networking: Linux, WebRTC, Android, Raspberry Pi, TCP/UDP sockets, multipath networking.

Development & analysis: Git, Docker, Wireshark, tcpdump, PyTorch, GNU Radio.

ENGINEERING & RESEARCH EXPERIENCE

University of Southern California / University of Minnesota

Ph.D. Student | Aug. 2022 - Present

- **Satellite network measurement, prediction, and optimization:** Collected and synchronized network/satellite data, preprocessed training inputs, and built PyTorch prediction models; contributed to Android measurement clients and a satellite-link prediction framework (HotMobile 2026). Co-first author of a 3,800+ km Starlink and 4G/5G measurement/MPTCP study (CoNEXT Companion 2023).
- **Multipath networking:** Built client, server, and proxy components using Raspberry Pi relay nodes; increased throughput by 50-80% and reduced packet-loss rate to about one-tenth of the single-satellite baseline.
- **WebRTC video-call optimization (INFOCOM 2026 Poster):** Built a WebRTC measurement system for satellite video calls using Docker; contributed to prediction-driven video transmission and reconstruction that adapts to link degradation.
- **NIER / low-bitrate neural video conferencing:** Contributed to rebuilding the neural model for Android phones; NIER reconstructs video from sparse keypoints and reference images.
- **Self-supervised network modeling:** Contribute to Net-JEPA, a 4M-parameter world model for adaptive streaming and congestion control; pretraining uses 450,000+ Puffer streams, with transfer calibrated on 20 traces.

SUTD | Research Assistant | May 2021-Apr. 2022

- Developed interpretable ML models for modulation classification and implemented an AI-based software modulator.

Southeast University | Graduate Researcher | Sept. 2019-Oct. 2020; Jun. 2021

- Designed and evaluated a multidimensional LoRa demodulator and implemented a fine-grained self-correlation collision detector (ICNP 2020; WASA 2021).

SELECTED PUBLICATIONS

Predicting Connectivity and Link Performance in Dynamic LEO Satellite Networks. **ACM HotMobile, 2026.** Bin Hu et al.

NN-Defined Modulator: Reconfigurable and Portable Software Modulator on IoT Gateways. **USENIX NSDI, 2024.** Jiazhao Wang et al. (co-author: Bin Hu)

INDUSTRY EXPERIENCE

Guangzhou Desheng Artificial Intelligence Technology Co. Ltd.

Algorithm Engineer (Intern) | Aug. 2018

- Developed camera-based assisted driving systems
- Implemented driver behavior detection and warning algorithms